nth Terms of Linear Sequences

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Written by: Mark Curtis
Reviewed by: Dan Finlay
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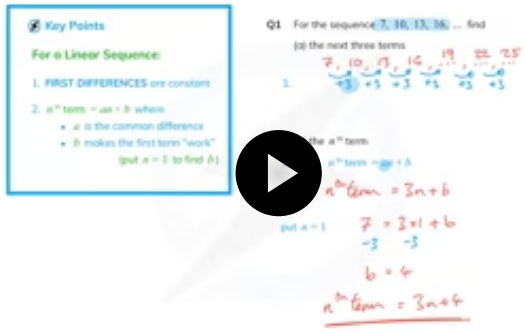

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Linear sequences

What is a linear sequence?

- A **linear** sequence goes **up** (or **down**) by the **same amount** each time
- This amount is called the **common difference, *d***
  - For example:
    1, 4, 7, 10, 13, ...(adding 3, so $d = 3$)
    15, 10, 5, 0, -5, ... (subtracting 5, so $d = -5$)
- Linear sequences are also called **arithmetic** sequences

How do I find the nth term formula for a linear sequence?

- The formula is ***n* th term = *dn* + *b***
  - ***d*** is the **common difference**
    - The amount it goes up by each time
  - ***b*** is the value **before** the **first** term (sometimes called the **zero** term)
    - Imagine going **backwards**
- For example 5, 7, 9, 11, ...
  - The sequence adds 2 each time
    - $d = 2$
  - Now continue the sequence backwards, from 5, by one term

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- For example 15, 10, 5, ...
  - Subtracting 5 each time means $d = -5$
  - Going backwards from 15 gives $15 + 5 = 20$
    - (20), 15, 10, 5, ... so $b = 20$
  - The n th term = $-5n + 20$

Worked Example

Find a formula for the nth term of the sequence -7, -3, 1, 5, 9, ...

Answer:

The n th term is $dn + b$ where d is the common difference and b is the term before the 1st term
The sequence goes up by 4 each time

$$d = 4$$

Continue the sequence backwards by one term (-7−4) to find b

$$(-11), -7, -3, 1, 5, 9, \ldots$$

$$b = -11$$

Substitute $d = 4$ and $b = -11$ into $dn + b$

$$n^{th} \text{ term} = 4n - 11$$

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